Receipt and Invoice Digitizer

OCR- Optical Character Recognition

- It is a technology that reads text from images

- It enables machines to recognize, extract, and digitize text, eliminating manual data entry, improving efficiency, and allowing for automation in industries like banking, healthcare.

How OCR works

- Takes an image
- Finds text areas in the image
- Recognizes each character
- Converts it into digital text

Chosen OCR Model: Paddle OCR

What is PaddleOCR?

PaddleOCR is an open-source Optical Character Recognition (OCR) toolkit developed by Baidu. It is designed to extract text from images with high accuracy using deep learning models.

What is PaddleOCR Engine?

- PaddleOCR is built on the PaddlePaddle deep learning framework
- It converts text in images into editable digital text
- It works well with scanned documents, photos, screenshots, receipts, and invoices
- Supports multiple languages and complex text layouts

- **Why PaddleOCR?**
- High accuracy on real-world images
- Handles noisy and low-quality images
- Preserves text layout and alignment
- Supports multiple languages

How Paddle OCR Works

How PaddleOCR Works

•Give an image to PaddleOCR

•PaddleOCR detects text regions in the image

•It recognizes characters using deep learning models

•Preserves layout and alignment

•Extracted text is returned to Python

What this code does

•Imports PaddleOCR library

•Initializes OCR model with English language

•Takes an image as input

•Extracts text from the image

•Prints the recognized text

1. **Install PaddleOCR**

```
pip install paddleocr
```

2.Install PaddlePaddle (CPU version)

```
pip install paddlepaddle
```

3.Import PaddleOCR in Python

```
from paddleocr import PaddleOCR
```

4. Initialize OCR model (English)

```
ocr = PaddleOCR(lang='en')
```

5. Text Extraction

```
result = ocr.ocr("image.jpg")
```

6. Print extracted text

```
for line in result:
    for word in line:
        print(word[1][0])
```

Pytesseract	PaddleOCR
Inconsistent decimal recognition	Accurate decimal recognition
Poor noise handling	Better layout preservation
Broken or misaligned text	High accuracy on receipts
Weak layout understanding	Handles multiple fonts well
Less reliable for receipts and complex documents	More reliable on real-world scanned images

DocTR (Document Text Recognition)	PaddleOCR
High accuracy on clean, well-scanned documents	Very high accuracy on real-world documents
Better suited for structured and simple layouts	Strong layout and text alignment preservation
Limited language support	Supports many languages
Sensitive to noise and low-quality images	Handles noisy images well
Less effective on complex layouts and receipts	Works well for receipts, invoices, and mixed fonts

SURYA OCR LIMITATIONS	PaddleOCR
Limited handwriting accuracy Struggles with cursive or messy handwriting	Better handwriting recognition compared to Surya OCR Handles cursive and informal writing more reliably
Accuracy drops on informal writing styles Slower for simple OCR tasks	Faster for simple and general OCR tasks Efficient even on small images
Takes more time on small images Not ideal when speed is a priority	Good balance between speed and accuracy
Complex installation and setup Not beginner-friendly for students Mainly suitable for basic text extraction, not optimized for speed or ease	Easier installation compared to Surya OCR Beginner-friendly and widely used Suitable for both basic and complex text extraction tasks

EasyOCR	PaddleOCR
Sensitive to image quality Blurry, low-resolution, or poorly scanned images reduce accuracy	More robust to image quality variations Performs better on blurry and low-resolution images
Low accuracy on noisy documents Shadows, glare, folds, and stains confuse OCR	Handles noisy documents more effectively Better tolerance to shadows, glare, and stains
Confuses similar characters Example: 0 vs O, 1 vs I	Lower character confusion compared to EasyOCR
Slower Than Traditional OCR EasyOCR uses deep learning models Processing time is slower than PyTesseract, especially on CPU	Faster processing than EasyOCR in most cases Better CPU and GPU performance More efficient resource usage Stronger results on low-quality images Better handwriting recognition than EasyOCR

THANK YOU

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